



BHARATIYA VIDYA BHAVAN'S
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Department of Computer Engineering

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AIM:	To install and configure Git, perform basic version control operations, work with remote repositories, implement branching strategies, collaborate using Git workflows, and handle merge conflicts effectively.
Program 1	
Part A:	Sudo apt update <pre>root@Shivsharan:/# apt install git -y Reading package lists... Done Building dependency tree... Done Reading state information... Done The following additional packages will be installed: adduser ca-certificates git-man krb5-locales less libbrotil1 libbsd0 libcbor0.10 libcurl3t64-gnutls libedit2 liberror-perl libexpat1 libfido2-1 libgdbm-compat4t64 libgdbm6t64 libgssapi-krb5-2 libk5crypto3 libkeyutils1 libkrb5-3 libkrb5support0 libldap-common libldap2 libnghttp2-14 libperl5.38t64 libpsl5t64 librtmp1 libsasl2-2 libsasl2-modules libsasl2-modules-db libssh-4 libx11-6 libx11-data libxau6 libxcb1 libxdmcp6 libxext6 libxmuu1 netbase openssh-client openssl patch perl perl-modules-5.38 publicsuffix xauth Suggested packages: liblocale-gettext-perl cron quota ecryptfs-utils gettext-base git-daemon-run git-daemon-sysvinit git-doc git-em git-gui gitk gitweb git-cvs git-mediawiki git-svn gdbm-l10n krb5-doc krb5-user libsasl2-modules-gssapi-mit libsasl2-modules-gssapi-heimdal libsasl2-modules-ldap libsasl2-modules-otp libsasl2-modules-sql keychain libpam-ssh monkeysphere ssh-askpass ed diffutils-doc perl-doc libterm-readline-gnu-perl libterm-readline-perl-p make libtap-harness-archive-perl</pre> sudo apt install git -y <pre>root@Shivsharan:/# apt install git -y Reading package lists... Done Building dependency tree... Done Reading state information... Done The following additional packages will be installed: adduser ca-certificates git-man krb5-locales less libbrotil1 libbsd0 libcbor0.10 libcurl3t64-gnutls libedit2 liberror-perl libexpat1 libfido2-1 libgdbm-compat4t64 libgdbm6t64 libgssapi-krb5-2 libk5crypto3 libkeyutils1 libkrb5-3 libkrb5support0 libldap-common libldap2 libnghttp2-14 libperl5.38t64 libpsl5t64 librtmp1 libsasl2-2 libsasl2-modules libsasl2-modules-db libssh-4 libx11-6 libx11-data libxau6 libxcb1 libxdmcp6 libxext6 libxmuu1 netbase openssh-client openssl patch perl perl-modules-5.38 publicsuffix xauth Suggested packages: liblocale-gettext-perl cron quota ecryptfs-utils gettext-base git-daemon-run git-daemon-sysvinit git-doc git-em git-gui gitk gitweb git-cvs git-mediawiki git-svn gdbm-l10n krb5-doc krb5-user libsasl2-modules-gssapi-mit libsasl2-modules-gssapi-heimdal libsasl2-modules-ldap libsasl2-modules-otp libsasl2-modules-sql keychain libpam-ssh monkeysphere ssh-askpass ed diffutils-doc perl-doc libterm-readline-gnu-perl libterm-readline-perl-p make libtap-harness-archive-perl</pre> git --version <pre>root@Shivsharan:/# git --version git version 2.43.0 root@Shivsharan:/#</pre>



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Part B:	Basic Git Configuration <pre>root@Shivsharan:/# git config --global user.name "Shivsharan" root@Shivsharan:/# git config --global user.email "shivsharan.sanjawad@gmail.com" root@Shivsharan:/# git config --global user.editor nano root@Shivsharan:/# git config --list user.name=Shivsharan user.email=shivsharan.sanjawad@gmail.com user.editor=nano root@Shivsharan:/#</pre>
Part C:	Step 1: Initialize Repository <pre>root@Shivsharan:/# mkdir git-lab root@Shivsharan:/# cd git-lab root@Shivsharan:/git-lab# git init hint: Using 'master' as the name for the initial branch. This default branch name hint: is subject to change. To configure the initial branch name to use in all hint: of your new repositories, which will suppress this warning, call: hint: hint: git config --global init.defaultBranch <name> hint: hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and hint: 'development'. The just-created branch can be renamed via this command: hint: hint: git branch -m <name> Initialized empty Git repository in /git-lab/.git/</pre> Step 2: Create File and Commit <pre>root@Shivsharan:/git-lab# echo "Hello from Shivsharan" > file.txt root@Shivsharan:/git-lab# git add file.txt root@Shivsharan:/git-lab# git commit -m "intial test" [master (root-commit) 0885df3] intial test 1 file changed, 1 insertion(+) create mode 100644 file.txt root@Shivsharan:/git-lab#</pre> Step 3: Check Status and Logs <pre>root@Shivsharan:/git-lab# git status On branch master nothing to commit, working tree clean root@Shivsharan:/git-lab# git log commit 0885df3403512c910e1ec76aa0e90d9da3f41463 (HEAD -> master) Author: Shivsharan <shivsharan.sanjawad@gmail.com> Date: Mon Mar 2 07:01:37 2026 +0000 intial test</pre>



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	<pre>root@Shivsharan:/git-lab# git diff diff --git a/file.txt b/file.txt index a9d230b..9efcd71 100644 --- a/file.txt +++ b/file.txt @@ -1,1 @@ -Hello from Shivsharan +Hello from Shivsharan to everyone root@Shivsharan:/git-lab# </pre>
Part D: Setting Up SSH Authentication	Step 1: Generate SSH Key <pre>root@Shivsharan:/git-lab# ssh-keygen -t ed25519 -C "shivsharan.sanjawad@gmail.com" Generating public/private ed25519 key pair. Enter file in which to save the key (/root/.ssh/id_ed25519): Created directory '/root/.ssh'. Enter passphrase (empty for no passphrase): Enter same passphrase again: Your identification has been saved in /root/.ssh/id_ed25519 Your public key has been saved in /root/.ssh/id_ed25519.pub The key fingerprint is: SHA256:78xfCUj22ABa83MgkfviQtef2AUe5kIpX1LRkwpF54 shivsharan.sanjawad@gmail.com The key's randomart image is: +--[ED25519 256]--+ =++o oo*==.. .X O*o.. . * Eo+Oo . + =S*o++ . o +. + . + . . +.. +-----[SHA256]-----+ root@Shivsharan:/git-lab# </pre> <pre>root@Shivsharan:~/ssh# ls -l total 8 -rw----- 1 root root 419 Mar 2 07:10 id_ed25519 -rw-r--r-- 1 root root 111 Mar 2 07:10 id_ed25519.pub root@Shivsharan:~/ssh# cat id_ed25519.pub ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAINwmGLtj/v4BC/IX8KXos6Rvsth6DTPUxjqw/TP5dVf shivsharan.sanjawad@gmail.com root@Shivsharan:~/ssh# </pre> <pre>root@Shivsharan:~/ssh# ls -l total 8 -rw----- 1 root root 419 Mar 2 07:10 id_ed25519 -rw-r--r-- 1 root root 111 Mar 2 07:10 id_ed25519.pub root@Shivsharan:~/ssh# cat id_ed25519.pub ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAINwmGLtj/v4BC/IX8KXos6Rvsth6DTPUxjqw/TP5dVf shivsharan.sanjawad@gmail.com root@Shivsharan:~/ssh# eval "\$(ssh-agent -s)" Agent pid 3271 root@Shivsharan:~/ssh# ssh -T git@github.com The authenticity of host 'github.com (20.207.73.82)' can't be established. ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU. This key is not known by any other names. Are you sure you want to continue connecting (yes/no/[fingerprint])? yes Warning: Permanently added 'github.com' (ED25519) to the list of known hosts. Hi ShivsharanSanjawad! You've successfully authenticated, but GitHub does not provide shell access. root@Shivsharan:~/ssh# </pre> <pre>root@Shivsharan:~/ssh# touch ~/.ssh/config root@Shivsharan:~/ssh# nano config root@Shivsharan:~/ssh# </pre>



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Part E:Working with Git	All steps of creating and adding files exactly same as done previously , now Directly pushing them to remote <pre>root@Shivsharan:/git-lab# git remote -v root@Shivsharan:/git-lab# git remote add origin git@github.com:ShivsharanSanjawad/gitLab.git root@Shivsharan:/git-lab# git branch -M main root@Shivsharan:/git-lab# git push -u origin main Enumerating objects: 3, done. Counting objects: 100% (3/3), done. Writing objects: 100% (3/3), 240 bytes 240.00 KiB/s, done. Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 To github.com:ShivsharanSanjawad/gitLab.git * [new branch] main -> main branch 'main' set up to track 'origin/main'. root@Shivsharan:/git-lab#</pre>



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```
From https://github.com/ShivsharanSanjawad/gitLab
c218c13..dedb442  main    -> origin/main
PS C:\Users\SHIVSHARN\Desktop\TEMP\gitLab> ls

Directory: C:\Users\SHIVSHARN\Desktop\TEMP\gitLab

Mode                LastWriteTime         Length Name
----                -
-a---             04-03-2026   23:21           93 file.txt
-a---             04-03-2026   23:21           44 README.md

PS C:\Users\SHIVSHARN\Desktop\TEMP\gitLab> git pull origin main
From https://github.com/ShivsharanSanjawad/gitLab
 * branch            main      -> FETCH_HEAD
Updating c218c13..dedb442
Fast-forward
 demoFile.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 demoFile.txt
PS C:\Users\SHIVSHARN\Desktop\TEMP\gitLab> ls

Directory: C:\Users\SHIVSHARN\Desktop\TEMP\gitLab

Mode                LastWriteTime         Length Name
----                -
-a---             04-03-2026   23:22           72 demoFile.txt
-a---             04-03-2026   23:21           93 file.txt
-a---             04-03-2026   23:21           44 README.md

PS C:\Users\SHIVSHARN\Desktop\TEMP\gitLab> |
```

**Part G: Branching
and Collaboration**



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	<pre>root@Shivsharan:/git-lab# git checkout -b newFeature Switched to a new branch 'newFeature' root@Shivsharan:/git-lab# echo "randomCode" >>file.txt root@Shivsharan:/git-lab# git add file.txt root@Shivsharan:/git-lab# git commti -m "newFeature added" git: 'commti' is not a git command. See 'git --help'. The most similar command is commit root@Shivsharan:/git-lab# git commit -m "newFeature added" [newFeature c1ba474] newFeature added 1 file changed, 2 insertions(+), 1 deletion(-) root@Shivsharan:/git-lab# git push origin newFeature Enumerating objects: 5, done. Counting objects: 100% (5/5), done. Writing objects: 100% (3/3), 293 bytes 293.00 KiB/s, done. Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 remote: remote: Create a pull request for 'newFeature' on GitHub by visiting: remote: https://github.com/ShivsharanSanjawad/gitLab/pull/new/newFeature remote: To github.com:ShivsharanSanjawad/gitLab.git * [new branch] newFeature -> newFeature root@Shivsharan:/git-lab#</pre> <pre>root@Shivsharan:/git-lab# git checkout main Switched to branch 'main' Your branch is up to date with 'origin/main'. root@Shivsharan:/git-lab# git merge newFeature Updating 0885df3..c1ba474 Fast-forward file.txt 3 ++- 1 file changed, 2 insertions(+), 1 deletion(-)</pre>
H: Branching Strategies	<pre>root@Shivsharan:/git-lab# git checkout main Switched to branch 'main' Your branch is up to date with 'origin/main'. root@Shivsharan:/git-lab# git merge newFeature Updating 0885df3..c1ba474 Fast-forward file.txt 3 ++- 1 file changed, 2 insertions(+), 1 deletion(-)</pre> <pre>c1ba474..d263393 newFeature -> newFeature root@Shivsharan:/git-lab# git checkout main Switched to branch 'main' Your branch is ahead of 'origin/main' by 2 commits. (use "git push" to publish your local commits) root@Shivsharan:/git-lab# git merge newFeature Auto-merging file.txt CONFLICT (content): Merge conflict in file.txt Automatic merge failed; fix conflicts and then commit the result. root@Shivsharan:/git-lab#</pre>



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	<pre>Hello from Shivsharan to everyone randomCode <<<<<< HEAD hello ===== What do you expect other than merge Conflict >>>>>> newFeature root@Shivsharan:/git-lab# git merge newFeature Auto-merging file.txt CONFLICT (content): Merge conflict in file.txt Automatic merge failed; fix conflicts and then commit the result. root@Shivsharan:/git-lab# nano file.txt root@Shivsharan:/git-lab# git add . root@Shivsharan:/git-lab# git commit -m "mergeConflict fixed" [main 754c6a2] mergeConflict fixed root@Shivsharan:/git-lab# Part J: Rebasing vs Merging root@Shivsharan:/git-lab# git rebase main Current branch main is up to date. root@Shivsharan:/git-lab# root@Shivsharan:/git-lab# git rebase main Current branch main is up to date. root@Shivsharan:/git-lab# cd .. root@Shivsharan:/# rm -rf git-lab root@Shivsharan:/# git config --global --unset-all user.name git config --global --unset-all user.email git config --global --unset-all core.editor root@Shivsharan:/# rm -rf ~/.ssh root@Shivsharan:/# eval "\$(ssh-agent -k)" Agent pid 3271 killed root@Shivsharan:/# history -c && history -w root@Shivsharan:/# </pre>
CONCLUSION:	Learned about the git and github. Learned how these work and how to use them for different purposes.